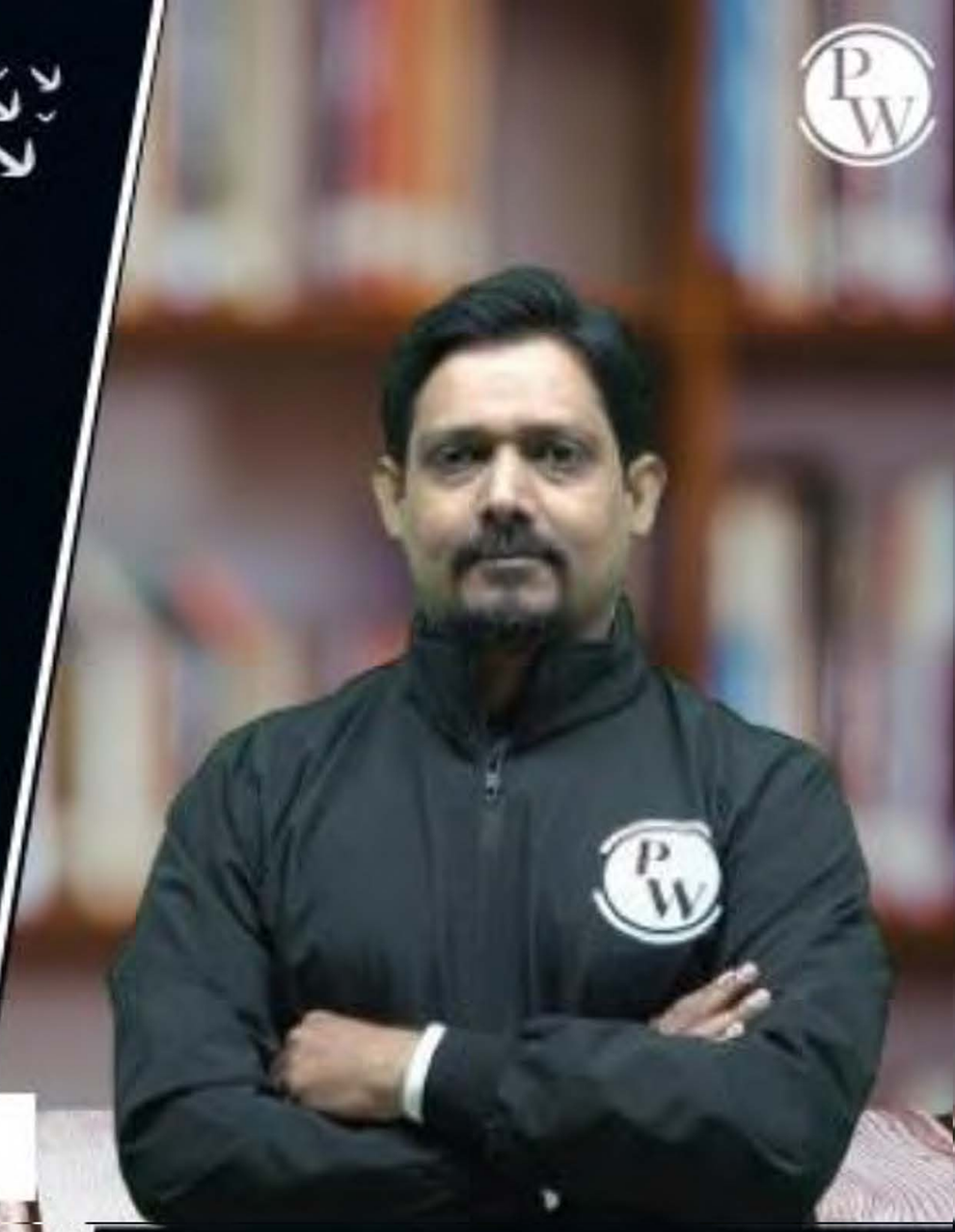
Computer Science & IT



Data Structure & Programming

Array

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

LTM - RMO

UTM - CMO

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

3-D array (C-Language)

Complexity

Slide



Topic : 3-D Matrix (C – Language)



Slide



3-D Array

a[1...3][1...3][1...3]

of Rows

of column

CD-ICG

frame

2-D entity

understand

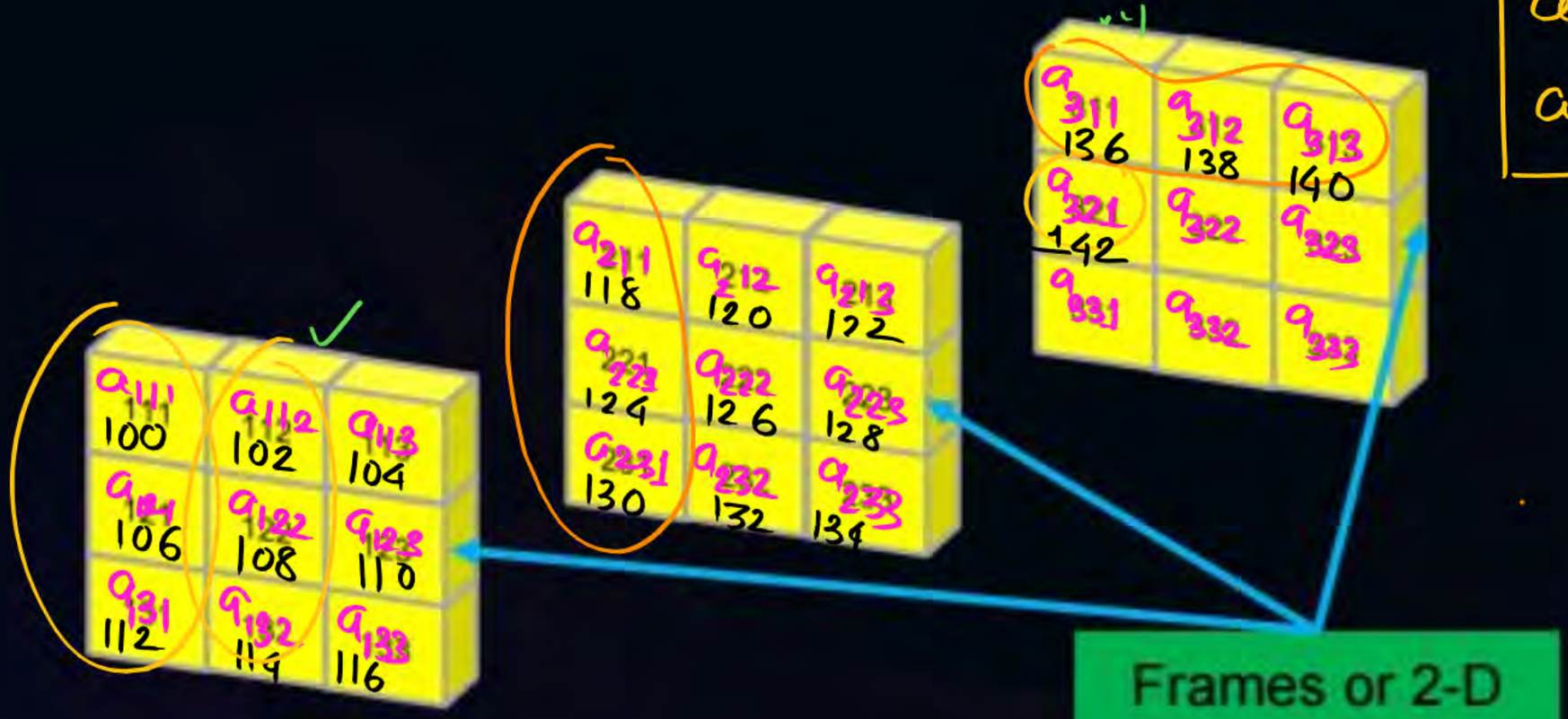






Frames or 2-D Array

3-D Array



a_{111}	a_{112}
a_{121}	a_{122}
a_{131}	a_{132}

Frames or 2-D Array



3-D Array

Lower bound - 1

a[3][2][1]

18 | +3+0

before 3rd frame — 2 frame complete

Size of frame 3×3 = 9 total. 2×9 = 18

---

on 3rd frame we are 2nd Row

(2-1) - 1 Row completed with 3 element total : 1×3 = 3

---

and 2nd Row, 1st column.

No. of column arranged (1-1) = 0

Slide



3-D Array

$$\text{total} = 18 + 3 + 0 = 21$$

Address of $a_{321} = 100 + 21 \times 2 = \boxed{142}$

Slide



3-D Array

$A[LB_1 \ldots UB_1][LB_2 \ldots UB_2][LB_3 \ldots UB_3]$

$A[i][j][k]$ BA base Address. Size = S

$$BA + \Big[(i - LB_1)(UB_2 - LB_2 + 1)(UB_3 - LB_3 + 1) + (j - LB_2) \times (UB_3 - LB_3 + 1) + (K - LB_3) \Big] \times S$$

Slide



Topic : 3-D Matrix (C – Language)

$A[1...100][1..100][1..100]$

BA - 100

Address $A[50][49][50]$

Size = 4B

$100 + [(50-1) \times 100 \times 100 + (49-1) \times 100 + (50-1)] \times 4$

$100 + [490000 + 4800 + 49] \times 4 = 100 + 494849 \times 4$

$= 100 + 1979396$

1979496

Slide



Topic : Question

#Q Consider the LTM consists of 45 non zero elements. If The dimension is n*n then the value of n is _____

Non zero

$1+2+3\ldots+n$

$\frac{n(n+1)}{2}$

$\frac{9\times 10}{2}=45$

$n=9$

$n=-10$ ✗

$(n+10)(n-9)=0$

$n^2+10n-9n-90$

n^2+n-90

$=0$

Slide



Topic : Question

1 2 3 4

#Q If natural number are stored in Array A[-10..10][20..30] Then the number stored in A[5][25] is

A[-10..10][20..30] RMO

while structure - logic

$$1 + [\underline{(5-(-10))} \times 11 + (25-20)] \times 1$$

$$1 + [165 + 5] = 170 + 1 = 171$$

Slide

$A[-10..10][20..30]$

$A[1...21][1..11]$

$A[5][25]$

$A[16][6]$



Topic : Question

#Q Consider the LTM a[-4..4][0...8].Non zero elements are stored in row major order. What is the address of A[0][3]. Base address is 1000 and size of each element is 2B.



Topic : Question

NAT

#Q A[1...100]1...100] is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space if the address of A[i][10] in column major order is 2808 then value of i is ____?

$$BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j) \right] \times S$$

$$1000 + \left[100 \times 9 - \frac{9 \times 8}{2} + (i-10) \right] \times 2 = 2808$$

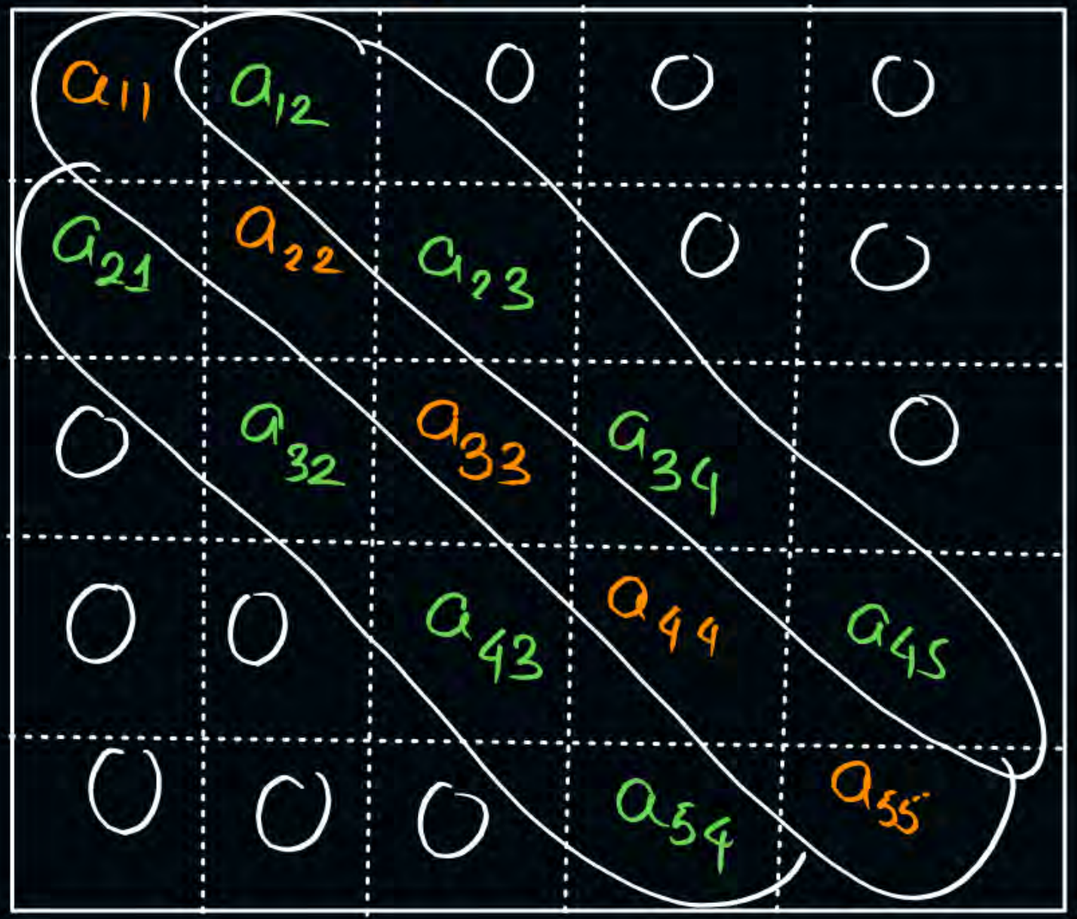
$$[900 - 36 + i - 10] \times 2 = 1808 \Rightarrow 900 - 46 + i = 904$$

$$i = 904 + 46 - 900$$

$$i = 50$$

Answer 50

Tridiagonal matrix

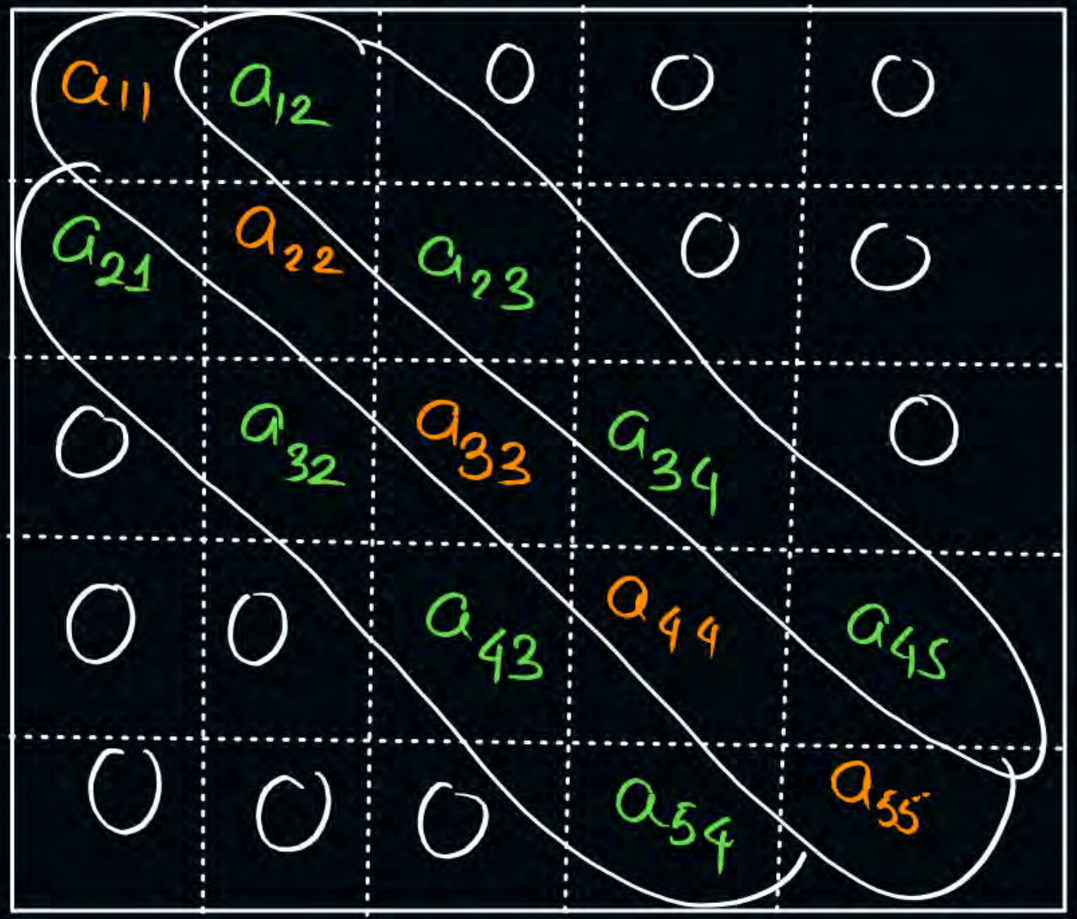


Non zero elements belongs

3 diagonal

$$n + n-1 + n-1 = 3n-2$$

Tridiagonal matrix



a_{ij}

$i-j=0$

$i-j=1$

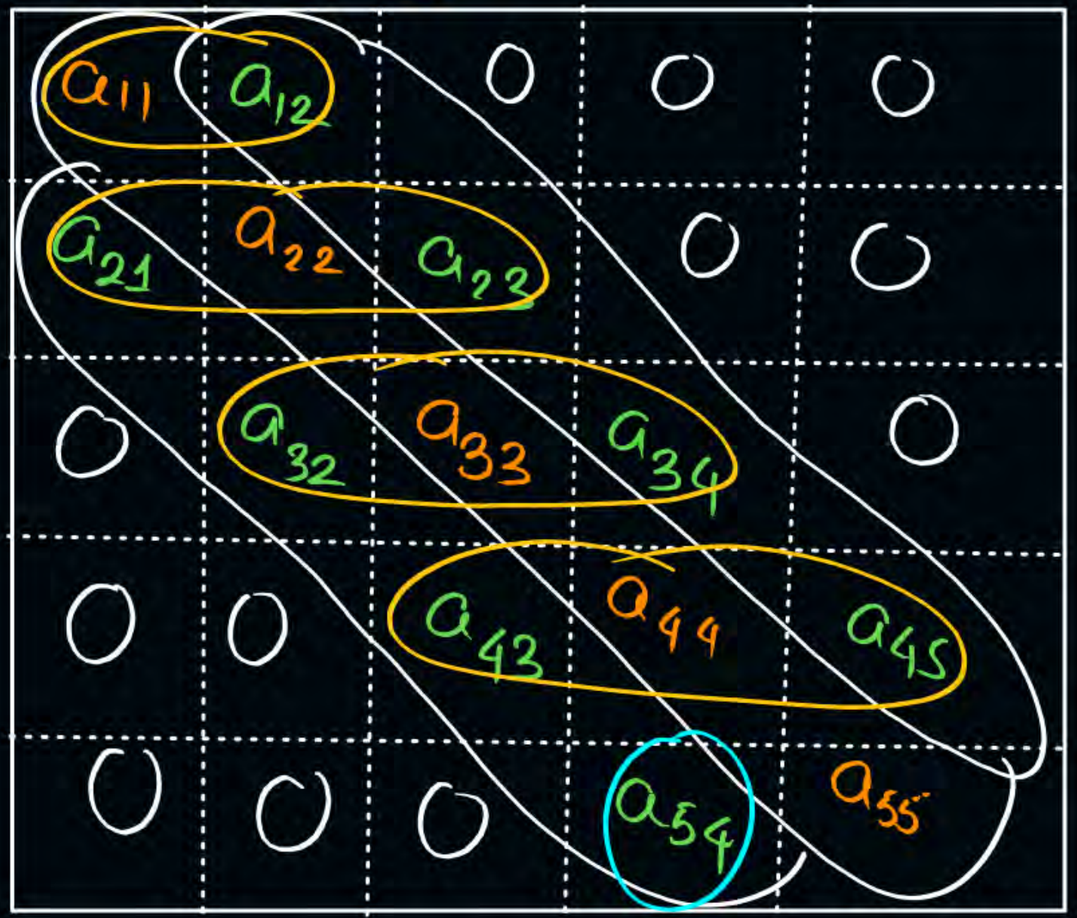
$j-i=1$

} Nonzero

RMO

a_{11}	a_{12}	a_{21}	a_{22}	a_{23}	a_{32}	a_{33}	a_{34}	a_{43}	a_{44}	a_{45}	a_{54}	a_{55}

Tridiagonal matrix



a_{ij}

i^{th} Row , $i-1$ Rows are completed

No. of element in $(i-1)$ Rows

$(i-1) \times 3 - 1$ ← first Row consist of 2 elements

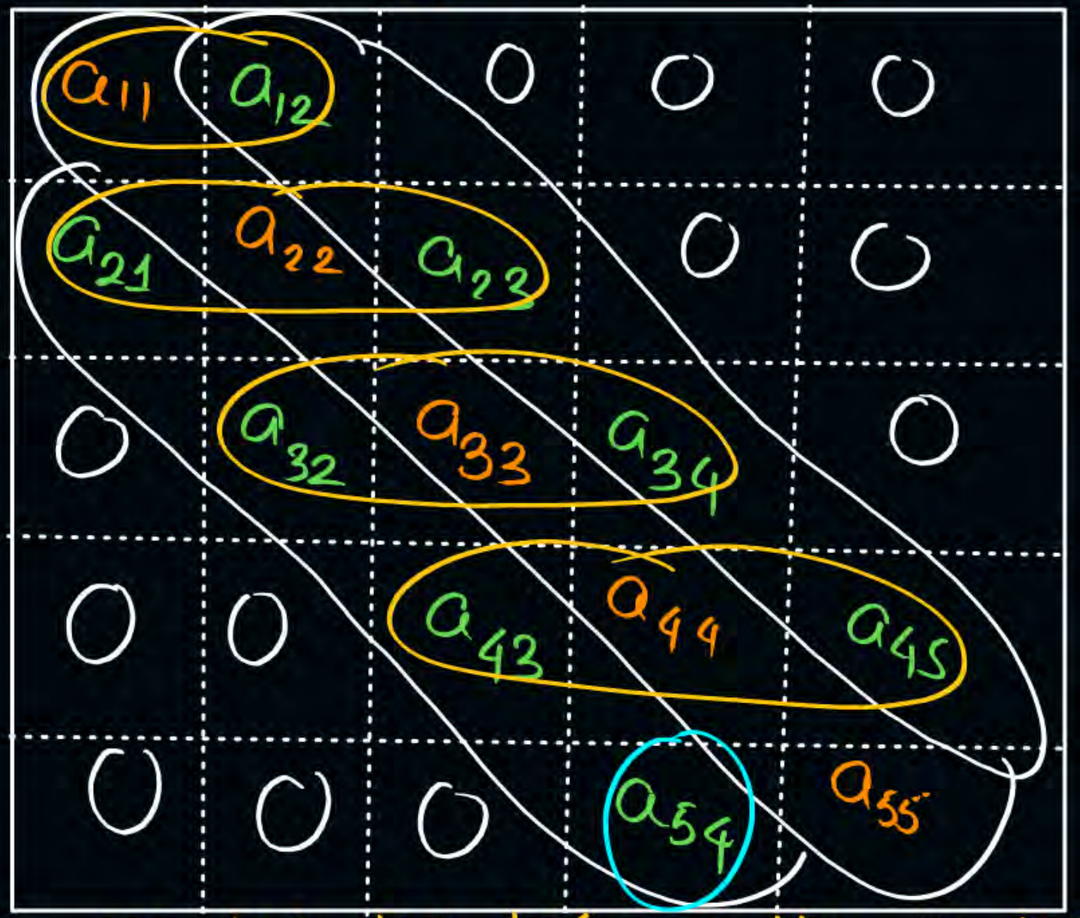
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i^{th} Row $\quad j - i + 1$

RMO

a_{11}	a_{12}	a_{21}	a_{22}	a_{23}	a_{32}	a_{33}	a_{34}	a_{43}	a_{44}	a_{45}	a_{54}	a_{55}

Tridiagonal matrix



$$BA + [(i-1)\times 3 - 1 + j - i + 1] \times S$$

$$BA + [3i - 3 + j - i] \times S$$

$$BA + [2i + j - 3] \times Size$$

$$100 + [10 + 4 - 3] \times 2$$

$$122$$

100	102	104	106	108	110	112	114	116	118	120	122	124
a_{11}	a_{12}	a_{21}	a_{22}	a_{23}	a_{32}	a_{33}	a_{34}	a_{43}	a_{44}	a_{45}	a_{54}	a_{55}

RMO

Tridiagonal column major - Homework

Data structure operation — Array, linked list

$O(n)$ — Size

depends upon Size

Search :- Array / linked

worst case

unordered last

$O(n)$

Approximation

max - 20 min

↑

O - Notation

Exact - Θ

min - Ω



find two element a_1, a_2

check every pair

$${}^{n}C_2 = \frac{n(n-1)}{2}$$

a_i a_j exists such that $a_i + a_j < 1000$

Best Algorithm

Independent of size

$O(1)$

Brute force



2 mins Summary

Topic

Topic

Topic

Topic

Topic

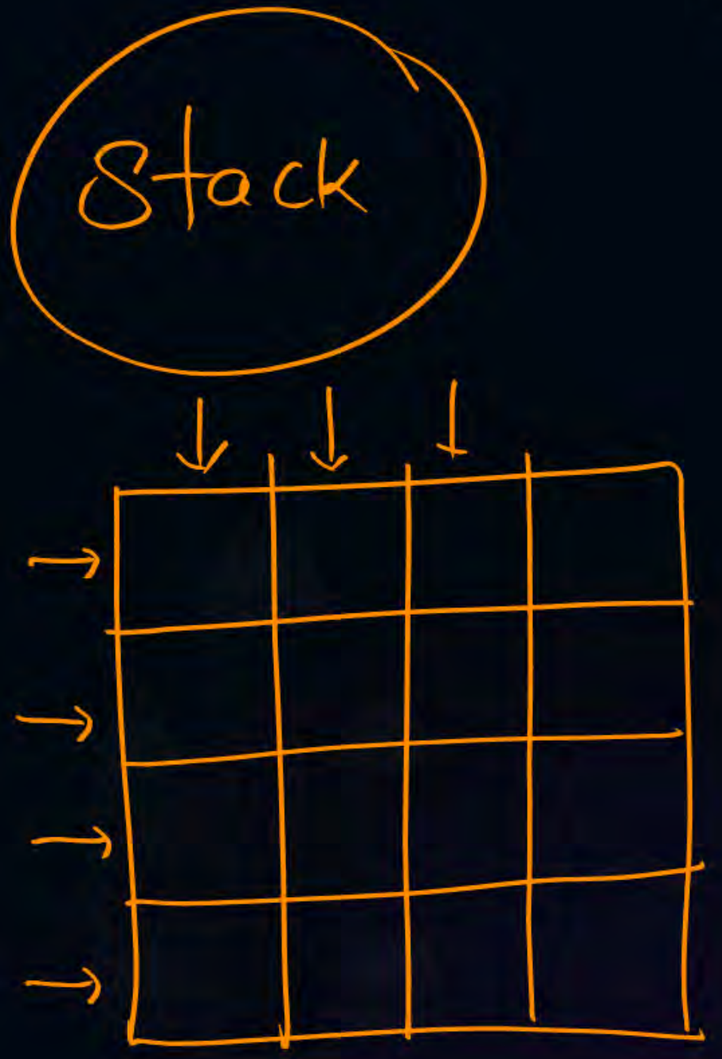
3-D array

Tridiagonal RMO

Complexity

Slide

THANK - YOU



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